

# Project: Restaurant Market Entry & Competitive Positioning Analysis

## A. Business context

**Business problem:** An entrepreneur is considering opening a new restaurant near Rodeo Drive, Beverly Hills. Before entering the market, the owner wants to understand how existing restaurants are positioned and identify potential opportunities for differentiation.

### **Business Objectives:**

Analyze the positioning of existing restaurants in the area and provide data-driven insights to support a preliminary market-entry decision.

### **Decision Questions:**

- How similar are existing restaurants in their market positioning?
- Which restaurants appear more differentiated from competitors?
- Where may there be room for a differentiated restaurant concept?
- What additional information should be considered before making a final investment decision?

**Stakeholder:** Prospective restaurant owner

**Data:** Public restaurant names and descriptions of 10 restaurants listed near Rodeo Drive on OpenTable.

**Tools:** Python, BeautifulSoup, Pandas, TF-IDF, Cosine Similarity, Seaborn

## B. Data Collection

**Purpose:** Public restaurant information was collected to build a structured view of the existing competitive landscape.

**Data Source:** Restaurant Overview from [Website](#)

**Data Crawling:**

## Skills: Webscraping, Data Preprocessing, Data Analysis, Data Visualization, Data Interpretation

```

21... from bs4 import BeautifulSoup
import urllib.request as req

21... url = "https://www.opentable.com/landmark/restaurants-near-rosario-drive"
html = req.urlopen(url)
doc = BeautifulSoup(html, "html.parser")

Visit the webpages of the 10 restaurants listed on the first page and crawl their names and descriptions

22... names = doc.find_all(class_="FhfgYo4tTD0-")
names

22... [<h6 class="FhfgYo4tTD0-">1. <!-- -->Ocean Prime - Beverly Hills</h6>,
<h6 class="FhfgYo4tTD0-">2. <!-- -->Jade Beverly Hills</h6>,
<h6 class="FhfgYo4tTD0-">3. <!-- -->The Hideaway</h6>,
<h6 class="FhfgYo4tTD0-">4. <!-- -->Negroni Beverly Hills</h6>,
<h6 class="FhfgYo4tTD0-">5. <!-- -->CUT by Wolfgang Puck at Beverly Wilshire</h6>,
<h6 class="FhfgYo4tTD0-">6. <!-- -->208 Rodeo - Beverly Hills</h6>,
<h6 class="FhfgYo4tTD0-">7. <!-- -->THE Blvd - Beverly Wilshire Hotel</h6>,
<h6 class="FhfgYo4tTD0-">8. <!-- -->Nua</h6>,
<h6 class="FhfgYo4tTD0-">9. <!-- -->The Grill on the Alley - Beverly Hills</h6>,
<h6 class="FhfgYo4tTD0-">10. <!-- -->Cipriani Beverly Hills</h6>]

22... names0 = [n.text for n in names]
names0

22... ['1. Ocean Prime - Beverly Hills',
'2. Jade Beverly Hills',
'3. The Hideaway',
'4. Negroni Beverly Hills',
'5. CUT by Wolfgang Puck at Beverly Wilshire',
'6. 208 Rodeo - Beverly Hills',
'7. THE Blvd - Beverly Wilshire Hotel',
'8. Nua',
'9. The Grill on the Alley - Beverly Hills',
'10. Cipriani Beverly Hills']

22... des = doc.find_all(class_="l9bbXUdC9v0- _3YSslEmZu4g- C7Tp-bANpE4- l-AMW5ZrIg-")

22... des0 = [d.text for d in des]
des0

22... ['Located just steps from Rodeo Drive, Ocean Prime is a nationally acclaimed, dynamic seafood & steakhouse from the award-winning Cameron Mitchell Restaurants. Ocean Prime delivers an elevated experience with exceptional ambiance, a chef-curated menu, signature handcrafted cocktails, Wine Spectator honored wine list and genuine hospitality. Reservations are recommended. Ocean Prime strives to provide an elegant, fine dining atmosphere. Our associates' priority is to ensure your experience is memorable. We ask that our guests choose attire that is both polished and suited to our restaurant.THANK YOU FOR NOT WEARING: Clothing emitting excessive marijuana odor Excessively revealing clothing or exposed undergarments Clothing with offensive language or graphics Admittance to Ocean Prime is at the discretion of management who reserves the right to refuse service to patrons who are inappropriately dressed. ',
'At Jade, we redefine Japanese cuisine in Beverly Hills, blending tradition with energy. Daytime offers quick, flavorful meals for professionals; nighttime transforms us into a hotspot with live music. Our menu fuses Japanese flavors with Western technique, featuring dishes like wagyu carpaccio with uni and white truffle oil. Try the wagyu donburi or the jewelry box bara chirashi for a rice-based delight. Indulge in the diamond roll with lobster, seared salmon, and caviar, or opt for the interactive D.

```

Figure 1. Webscraping Code

| Restaurant  | Restaurant Description                                                                                                                                                                                                                    |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ocean Prime | Located just steps from Rodeo Drive, Ocean Prime is a nationally acclaimed, dynamic seafood & steakhouse from the award-winning Cameron Mitchell Restaurants. Ocean Prime delivers an elevated experience with exceptional ambiance,..... |

|                    |                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jade Beverly Hills | At Jade, we redefine Japanese cuisine in Beverly Hills, blending tradition with energy. Daytime offers quick, flavorful meals for professionals, nighttime transforms us into a hotspot with live music..... |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table 1. Data Sample

### C. Data Preprocessing & Similarity Analysis

**Data Preprocessing:** Restaurant descriptions contain common words that do not meaningfully distinguish positioning. The text was standardized by lowercasing, removing stopwords, and stemming before comparison

Data Preprocessing: Use TF-IDF scores, convert upper case letters to lower case, delete stopwords, and stem the words.

```
from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords
from nltk.stem import PorterStemmer

for d in des0:
    d1 = word_tokenize(d)
    d2 = [w.lower() for w in d1 if w.isalpha()]
    d3 = [w for w in d2 if not w in new_stopwords]
    d4 = [ps.stem(w) for w in d3]
    print(d4)

['locat', 'step', 'rodeo', 'drive', 'ocean', 'prime', 'nation', 'acclaim', 'dynam', 'seafood', 'steakh',
ous', 'cameron', 'mitchel', 'restaur', 'ocean', 'prime', 'deliv', 'elev', 'experi', 'except', 'ambian',
c', 'menu', 'signatur', 'handcraft', 'cocktail', 'wine', 'spectat', 'honor', 'wine', 'list', 'genuin',
'hospit', 'reserv', 'recommend', 'ocean', 'prime', 'strive', 'provid', 'eleg', 'fine', 'dine', 'atmosp',
her', 'associ', 'prioriti', 'ensur', 'experi', 'memor', 'ask', 'guest', 'choos', 'attir', 'polish', 's',
uit', 'wear', 'cloth', 'emit', 'excess', 'marijuana', 'odor', 'excess', 'reveal', 'cloth', 'expos', 'u',
ndergar', 'cloth', 'offens', 'languag', 'graphic', 'admitt', 'ocean', 'prime', 'discret', 'manag', 're',
serv', 'right', 'refus', 'servic', 'patron', 'inappropri', 'dress']

['jade', 'redefin', 'japanes', 'cuisin', 'beverli', 'hill', 'blend', 'tradit', 'energi', 'daytim', 'of',
fer', 'quick', 'flavor', 'meal', 'profession', 'nighttim', 'transform', 'hotspot', 'live', 'music', 'm',
enu', 'fuse', 'japanes', 'flavor', 'western', 'techniqu', 'featur', 'dish', 'like', 'wagyu', 'carpacci',
o', 'uni', 'white', 'truffl', 'oil', 'tri', 'wagyu', 'donburi', 'jewelri', 'box', 'bara', 'chirashi',
'delight', 'indulg', 'diamond', 'roll', 'lobster', 'sear', 'salmon', 'caviar', 'opt', 'interact', 'han',
d', 'roll', 'platter', 'elev', 'experi', 'caviar', 'servic', 'carat', 'premium', 'dragon', 'ultim', 'g',
roup', 'celebr', 'unforgett', 'culinari', 'journey', 'craft', 'delight', 'sens']

['hidden', 'insid', 'rodeo', 'collect', 'heart', 'beverli', 'hill', 'hideaway', 'featur', 'beauti', 'o',
utdoor', 'patio', 'intim', 'indoor', 'privat', 'cabana', 'exquisit', 'bar', 'culinari', 'journey', 'fi',
ll', 'rich', 'vibrant', 'flavor', 'mexico', 'complement', 'handcraft', 'margarita']

['negrioni', 'caff', 'bar', 'well', 'renown', 'brand', 'serv', 'actual', 'locat', 'miami', 'bueno', 'ai',
r', 'punta', 'del', 'est', 'montevideo', 'asuncion', 'paraguay', 'beverli', 'hill', 'locat', 'featur',
'divers', 'intern', 'comfort', 'food', 'menu', 'offer', 'appet', 'salad', 'entre', 'burger', 'brunch',
'combin', 'sushi', 'bar', 'offer', 'special', 'varietti', 'roll', 'nigiri', 'sashimi', 'crispi', 'ric',
e', 'special', 'mention', 'cocktail', 'bar', 'stand', 'serv', 'great', 'craft', 'cocktail', 'homemad',
'varietti', 'nearoni', 'spritz']
```

```
def my_tokenizer(d):
    d1 = word_tokenize(d)
    d2 = [w.lower() for w in d1 if w.isalpha()]
    d3 = [w for w in d2 if not w in new_stopwords]
    d4 = [ps.stem(w) for w in d3]
    return (d4)
```

Figure 2. Preprocessing Code

## Similarity Analysis (TF-IDF + Cosine similarity)

- Why TF-IDF?: TF-IDF converts restaurant descriptions into numerical representations while giving greater weight to more distinctive terms.
- Why Cosine Similarity? Cosine similarity measures how similar the restaurant descriptions are in terms of their textual positioning.

### Cosine Similarity

```
from sklearn.feature_extraction.text import TfidfVectorizer

tfidf = TfidfVectorizer (tokenizer = my_tokenizer, token_pattern = None)

tfidf_des0 = tfidf.fit_transform(des0)

tfidf_des0.toarray()

array([[0.         , 0.07526443, 0.         , ..., 0.         , 0.         ,
        0.         ],
       [0.         , 0.         , 0.         , ..., 0.         , 0.         ,
        0.         ],
       [0.         , 0.         , 0.         , ..., 0.         , 0.         ,
        0.         ],
       ...,
       [0.         , 0.         , 0.         , ..., 0.         , 0.         ,
        0.38354077],
       [0.         , 0.0755682 , 0.         , ..., 0.         , 0.20321421,
        0.         ],
       [0.         , 0.         , 0.         , ..., 0.09914401, 0.         ,
        0.         ]])

from sklearn.metrics.pairwise import cosine_similarity

cos_sim=cosine_similarity(tfidf_des0)

cos_sim

array([[1.         , 0.02751066, 0.02533322, 0.02502708, 0.04301644,
        0.10415201, 0.06908396, 0.01368665, 0.1953677 , 0.02864652],
       [0.02751066, 1.         , 0.08930411, 0.06160761, 0.04581626,
        0.00771906, 0.05536842, 0.         , 0.05129499, 0.0348296 ],
       [0.02533322, 0.08930411, 1.         , 0.06074112, 0.04433709,
        0.         , 0.17685099, 0.         , 0.04328569, 0.04732086],
       [0.02502708, 0.06160761, 0.06074112, 1.         , 0.07210486,
        0.         , 0.13637662, 0.03581288, 0.10753879, 0.0882945 ],
       [0.04301644, 0.04581626, 0.04433709, 0.07210486, 1.         ,
        0.00575576, 0.08543889, 0.0470263 , 0.0866316 , 0.05545376],
       [0.10415201, 0.00771906, 0.         , 0.         , 0.00575576,
        1.         , 0.07553551, 0.         , 0.0487256 , 0.05384368],
       [0.06908396, 0.05536842, 0.17685099, 0.13637662, 0.08543889,
        0.07553551, 1.         , 0.05489164, 0.10649934, 0.11489407],
       [0.01368665, 0.         , 0.         , 0.03581288, 0.0470263 ,
        0.         , 0.05489164, 1.         , 0.         , 0.         ],
       [0.1953677 , 0.05129499, 0.04328569, 0.10753879, 0.0866316 ,
        0.0487256 , 0.10649934, 0.         , 1.         , 0.09090357],
       [0.02864652, 0.0348296 , 0.04732086, 0.0882945 , 0.05545376,
        0.05384368, 0.11489407, 0.         , 0.09090357, 1.         ]])
```

Figure 3. Analysis code

**Result Interpretation:** Higher similarity suggests greater overlap in how restaurants position themselves, lower similarity suggests more differentiated positioning.

## D. Competitive Landscape Data Visualization

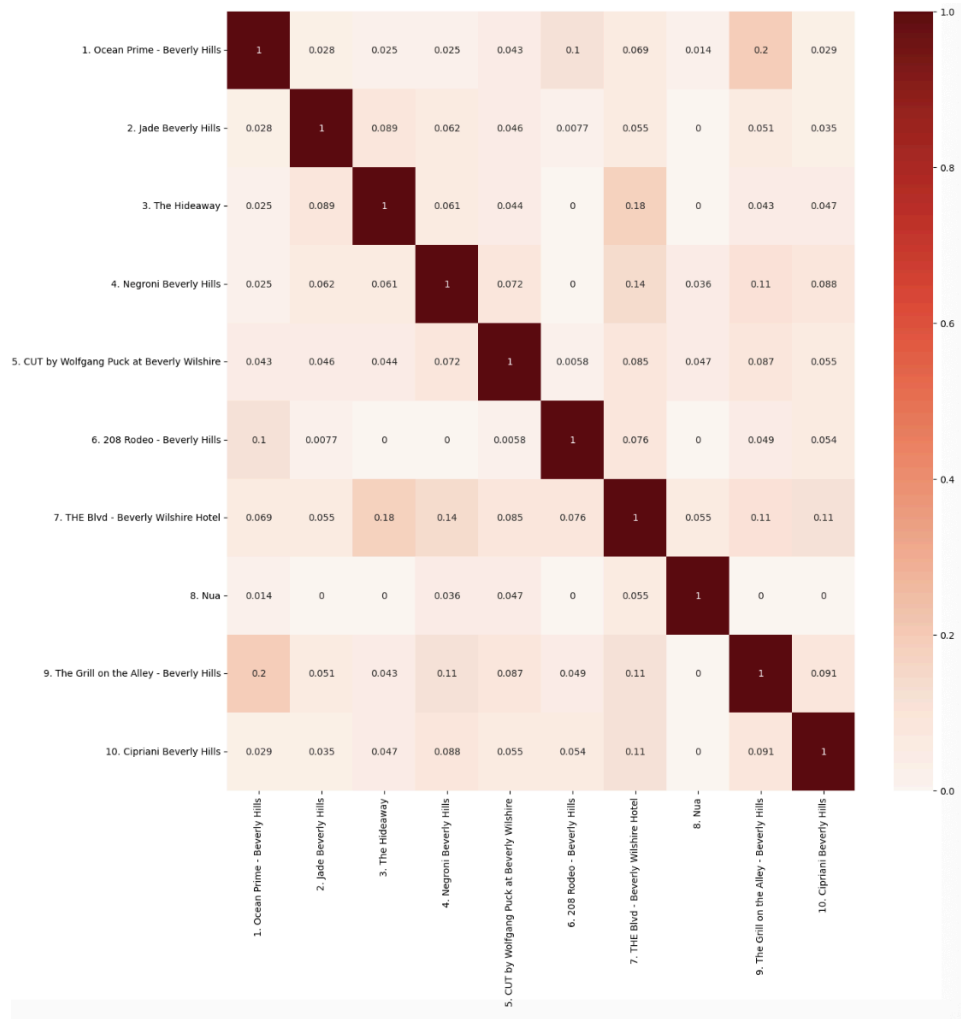


Figure 4. Similarity (Heatmap Visualization)

## Competitive Positioning Analysis

- Ocean Prime and The Grill on the Alley show the highest observed similarity in the sample (0.20), followed by The Hideaway and The Blvd (0.18).

## Key findings:

- Differentiated positioning: Nua shows very low similarity with most of the other restaurants in the sample, suggesting relatively differentiated textual positioning.

- **Competitive overlap:** Several restaurant pairs show higher similarity scores, indicating greater overlap in positioning and potential direct competition.

### **E. Business Recommendation**

**Business Interpretation:** The analysis suggests that some restaurants occupy relatively similar positioning spaces, while others, such as Nua, appear more differentiated based on their public descriptions.

#### **Recommendations:**

- **Avoid direct imitation:** A new entrant should avoid simply replicating concepts that already show strong positioning overlap with existing restaurants.
  - **Explore differentiation:** The owner should explore differentiation through cuisine, atmosphere, service concept, target customer occasion, or overall dining experience
  - **Validate demand before investment:** Low similarity alone does not indicate an attractive market opportunity. A differentiated concept should be validated against actual customer demand before launch. In addition, additional data should be evaluated, including customer ratings and reviews, cuisine category, average price level,...
- 
- **Note:** Please find the Full Coding File.ipynb attached